Column: Quantum Technologies

Quantum in Consumer Technology

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THE NEW QUANTUM technology is a rapidly developing field that has the potential to revolutionize a wide range of industries, including consumer technology. One of the most exciting current applications of quantum technology in the consumer industry is the area of quantum computing. Quantum computers use quantum bits, or qubits, instead of classical bits, which allows them to perform certain calculations much faster than traditional computers. This has the potential to greatly improve fields such as machine learning, drug discovery, financial modeling, and cryptography.

One example of a company developing quantum computers is IBM. IBM's quantum efforts were very close to the consumer industry in recent years. I remember IBM presenting the world's largest quantum computer at CES 2018 [see Figure 1(a)]. The machine was not operating at the show, but it was present at the Innovation Awards Pavilion. One year later, at CES 2019, the quantum computer was at the IBM booth [see Figure 1(b)]. By that time, I met Robert Loredo, IBM's Master Inventor and Quantum Ambassador. I recommend any of the readers interested in developing a quantum computing initiative in-house, contact Robert. You won't regret it. At CES 2020, anyone entering the show through the main entrance of the Central Hall of the Las Vegas Convention Center would be welcomed by IBM's claim "Enter the Quantum Decade" [see Figure 1(c)].

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QUANTUM DECADE

We are, in fact, in the quantum decade. The ability of quantum computers to process a large amount of data quickly and accurately can enable new and improved products and services for consumers.

In the area of machine learning, for example, quantum algorithms can be used to greatly speed up the training of machine learning models, allowing for faster and more accurate predictions. This has the potential to improve a wide range of consumer applications, such as personal assistants, image and speech recognition, and natural language processing.

Another important use of quantum computing in consumer technology is in drug discovery. Quantum computing can be used to simulate the behavior of molecules, which can greatly speed up the process of discovering new drugs. This has

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(a) (b) (c)

FIGURE 1. (a) During CES 2018, IBM showed "the world's largest quantum computer" at the Innovation Awards Pavilion. (b) IBM Q System One, Robert Loredo (right), IBM quantum ambassador, with the author at CES 2019. (c) Main entrance at CES 2020 of the Central Hall of the Las Vegas Convention Center with IBM Quantum Machine (pictures by the author).

the potential to improve the treatment of a wide range of diseases and improve people's health.

As said, quantum computing can be used to improve financial modeling. It can be used to simulate the behavior of financial markets and to optimize investment strategies. This allows the improvement of the efficiency of the financial industry.

Quantum computing can also be used for cryptography, for example, for improving the security of online transactions. This can be achieved by new quantum algorithms, used in encryption keys.

Beyond quantum computing, another potential application of quantum technology in consumer technology is in quantum sensors. These sensors use the properties of quantum mechanics to measure things such as temperature, pressure, and magnetic fields. This has the potential to greatly improve the performance of a wide range of consumer devices and supporting technologies, including more accurate positioning systems or brain imaging tools. Quantum teleportation allows the transfer of information between two points. This has applications in communication systems.

QUANTUM AND THE METAVERSE

Quantum may have a role in the metaverse, the virtual reality space where users can interact with each other and digital objects in a shared environment.

To improve the user experience in virtual reality games and social environments, it is necessary to create more realistic and immersive virtual worlds by simulating the behavior of large numbers of objects and characters in real time. A large amount of computing power is necessary for this task. Quantum computing might be useful here. The ability of quantum computers to process large amounts of data quickly and accurately can be used to create more efficient and scalable metaverse platforms. This can enable more users to participate in a virtual world simultaneously, and can also enable the creation of more complex and detailed virtual environments.

Quantum sensors might be used in future devices to improve accuracy in virtual reality experiences.

Quantum computing can also be used to improve the security of the metaverse. Quantum encryption can be used to secure communication between users and to protect the integrity of virtual environments. This can greatly improve the security of virtual transactions and can also help to prevent fraud and hacking.

OPTIMIZATION PROBLEMS

One of the main applications of quantum computing is combinatorial optimization. Some manufacturers develop machines, called "quantum annealers" or "digital annealers,"

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which size allows the approach to some real-world problems today.

Combinatorial optimization is a branch of mathematics that deals with finding the best solution to problems that involve a large number of discrete choices and may include restrictions. In general, large problems involve many variables and restrictions. These types of problems can be found in many different areas of the consumer industry and can have a significant impact on the efficiency and effectiveness of various processes.

One example of a large combinatorial optimization problem in the consumer industry is the problem of scheduling production in a manufacturing facility. This problem involves determining the best order to produce many products, taking into account factors such as the availability of resources, the lead time, or the priority of each product.

Another large combinatorial optimization problem in the consumer industry is the problem of routing delivery vehicles. This problem involves determining the best route for a large number of vehicles to take to deliver products to customers, taking into account factors such as traffic, road closures, and delivery times. Similarly, bandwidth allocation for cellular mobile and radio stations can be approached with these techniques. These problems can have a significant impact on the cost and efficiency of delivery or resources employed.

In retail, a combinatorial optimization problem is inventory management. This problem involves determining the best way to stock products in a store, taking into account factors such as customer demand, the cost of each product, and the storage space available. This problem can be very complex and can significantly impact a store's sales and profitability.

Finally, in the e-commerce or media consumption areas, the problem of combinatorial optimization can be seen in the problem of product recommendations. This problem involves determining the best products to recommend to a customer based on their purchase history, browsing history, and other factors. It can influence customer satisfaction.

These are just a few examples of the large combinatorial optimization problems that can be found in the consumer industry. These types of problems can be very complex and can have a significant impact on the efficiency, effectiveness, and cost of production, delivery, and other processes. By using mathematical techniques and optimization algorithms, companies can find the best solutions to these problems and improve their operations. Quantum computing has much to give in efficiently solving this kind of problem.

CTSOC AND QUANTUM TECHNOLOGY

CTSoc has established its *Quantum in Consumer Technology Technical Committee* (QCT TC). This committee, although new, has been organizing and sponsoring some activities and discussions.

CTSoc is a technical sponsor of IEEE Quantum Week, which is a multidisciplinary quantum computing and engineering venue that gives attendees the unique opportunity to discuss challenges and opportunities with quantum researchers, scientists, engineers, entrepreneurs, developers, students, practitioners, educators, programmers, and newcomers. In 2022, the event had more than 1 000 registrants. During this event, CTSoc organized a Workshop on Quantum in Consumer Technology.

Additionally, CTSoc has had quantum computing in its distinguished lecturer series.

In the recent ICCE 2023, there has been a Quantum Computing & Simulation session. More on the field is coming in other CTSoc events. Besides, CTSoc has sponsored other events such as Quantum Latino in 2021 and 2022 (https://quantum-latino.com/).

Participation in the QCT TC is open for IEEE members and nonmembers, as in other CTSoc's TCs. Join us at https://ctsoc.ieee.org/technical/technical-committees/qct-tc.html.

CONCLUSION AND THE FUTURE

Quantum technology is a rapidly developing field that has the potential to revolutionize a wide range of industries, including consumer technology. Current applications include quantum computing and quantum encryption, while future use cases include quantum sensors and quantum teleportation. Shortly, it is expected that more and more companies will begin to use quantum computing to improve their products

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and services. This includes companies in a wide range of industries, including finance, healthcare, and technology. As technology continues to evolve, it will be exciting to see how it will change the way we live and interact with technology. Join CTSoc and its Quantum in Consumer Technology Technical Committee to build the industry's future.

ABOUT THIS COLUMN

This is the first of a series of columns about quantum technologies in which we will develop the ideas outlined in this article and more news about the developments and applications of quantum technologies in the consumer technology industry and services, and about the companies working in the field as providers of hardware or software, or as final users.

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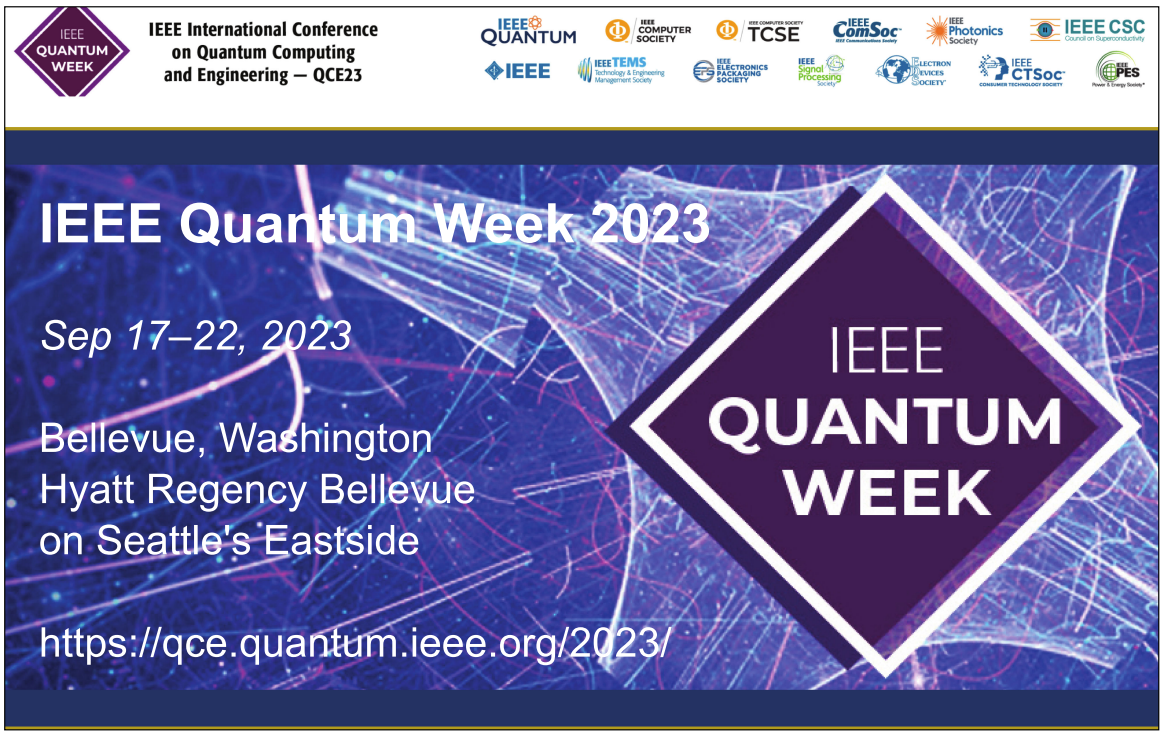

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